

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:	25/08/2026	Duration:	60 minutes

Code of Conduct

I Sk. Parvez Ahamed (192411084) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sk. Parvez Ahamed

Annexure A – Test Case Design Assignment

Blockchain-Based Academic Credential Verification Platform

1. Introduction

The Blockchain-Based Academic Credential Verification Platform is a digital platform designed to issue, store, manage, and verify academic credentials using blockchain technology. Educational institutions can issue credentials such as degrees, diplomas, and certificates to students. The credential information is securely recorded on the blockchain, providing integrity and resistance to unauthorized modification. Students can access and share their credentials, while employers, universities, and other authorized organizations can verify the authenticity of the credentials using a unique credential ID or QR code.

The objective of testing is to ensure that the platform performs its intended functions correctly, protects credential information, detects unauthorized modifications, and provides reliable and user-friendly verification.

2. Functional and Non-Functional Requirements

Type	Requirement ID	Requirement to be Tested
Functional	FR-01	Users should be able to register and log in to the platform.
Functional	FR-02	Authorized educational institutions should be able to issue academic credentials.
Functional	FR-03	The system should validate student and credential information before issuance.
Functional	FR-04	Credential records or hashes should be stored on the blockchain.
Functional	FR-05	The system should generate a unique credential ID and QR code.
Functional	FR-06	Students should be able to view and access their credentials.
Functional	FR-07	Authorized users should be able to verify academic credentials.
Functional	FR-08	The system should detect modified or tampered credentials.
Functional	FR-09	Authorized institutions should be able to revoke credentials.
Functional	FR-10	The system should display the correct status of revoked credentials.
Functional	FR-11	Unauthorized users should not be allowed to issue or revoke credentials.
Functional	FR-12	Users should be able to log out and sessions should be properly managed.
Non-Functional	NFR-01	The system should provide secure authentication and authorization.
Non-Functional	NFR-02	Credential data should maintain integrity and consistency.
Non-Functional	NFR-03	Credential verification should provide a reasonable response time.
Non-Functional	NFR-04	The system should support multiple users and verification requests.

Type	Requirement ID	Requirement to be Tested
Non-Functional	NFR-05	The system should handle blockchain or network failures gracefully.
Non-Functional	NFR-06	The interface should be simple and easy for users to understand.
Non-Functional	NFR-07	The platform should work correctly on supported browsers and devices.

3. Test Scenarios

The following scenarios are identified from the major functions of the platform.

Scenario ID	Test Scenario	Type
TS-01	Register a new user with valid and invalid information.	Functional
TS-02	Login using valid, invalid, and incomplete credentials.	Functional
TS-03	Issue an academic credential using valid student information.	Functional
TS-04	Attempt credential issuance with missing or invalid information.	Functional
TS-05	Store the credential record/hash on the blockchain.	Functional
TS-06	Generate and use a unique credential ID and QR code.	Functional
TS-07	View and access an issued academic credential.	Functional
TS-08	Verify a valid academic credential using ID or QR code.	Functional
TS-09	Verify a non-existent or invalid credential.	Functional
TS-10	Detect a credential whose information has been modified.	Functional/Security
TS-11	Revoke an academic credential and verify its revoked status.	Functional
TS-12	Prevent unauthorized users from issuing or revoking credentials.	Security

Scenario ID	Test Scenario	Type
TS-13	Handle blockchain or network failure during credential operations.	Exceptional
TS-14	Test logout and expired sessions.	Security
TS-15	Test response time, usability, and multiple verification requests.	Non-Functional

4. Detailed Test Cases

Test Case ID	Test Scenario	Test Steps / Test Data	Expected Result	Actual Result	Pass/Fail
TC-01	User Registration	Open registration page → enter valid name, email and password → submit. Data: parvezsk6@gmail.com, Secure@123	User account should be created successfully and confirmation should be displayed.	—	Pass
TC-02	Invalid Registration	Enter invalid email, weak password, or leave mandatory fields empty. Data: parvez@, blank name	Registration should be rejected and appropriate validation messages should be displayed.	—	Pass
TC-03	Valid Login	Enter registered email and correct password.	User should be successfully authenticated and redirected to the appropriate dashboard.	—	Pass
TC-04	Invalid Login	Enter correct email with incorrect password or blank credentials.	Login should fail and an appropriate error message should be displayed.	—	Pass

Test Case ID	Test Scenario	Test Steps / Test Data	Expected Result	Actual Result	Pass/Fail
TC-05	Valid Credential Issuance	Authorized institution logs in → enters student name, ID, degree and graduation year → submits. Data: Parvez, STU2026001, B.Tech CSE, 2026	Credential should be created successfully.	—	Pass
TC-06	Invalid Credential Issuance	Leave required information blank or enter invalid student ID/year.	System should prevent issuance and display validation errors.	—	Pass
TC-07	Credential ID & QR Generation	Successfully issue a credential and open the generated credential.	A unique credential ID and QR code should be generated and associated with the correct credential.	—	Pass
TC-08	Blockchain Storage	Issue a valid credential and verify the blockchain transaction/record.	Credential hash/record should be stored successfully and remain consistent.	—	Pass
TC-09	View Credential	Student logs in and opens the credentials section.	Student should be able to view the correct academic credential details.	—	Pass
TC-10	Valid Credential Verification	Enter an existing credential ID or scan its QR code. Data: CRD20260001	System should retrieve the correct credential and display it as authentic/valid.	—	Pass
TC-11	Invalid Credential Verification	Enter a non-existent or incorrectly formatted credential ID. Data: CRD999999	System should reject the request and display “Credential Not Found” or	—	Pass

Test Case ID	Test Scenario	Test Steps / Test Data	Expected Result	Actual Result	Pass/Fail
			equivalent message.		
TC-12	Tampered Credential Verification	Modify credential information after issuance and attempt verification.	System should detect the mismatch/hash difference and identify the credential as invalid or tampered.	—	Pass
TC-13	Credential Revocation	Authorized institution selects an active credential and chooses revoke.	Credential should be successfully revoked and its status should change to revoked.	—	Pass
TC-14	Verify Revoked Credential	Attempt to verify a previously revoked credential.	System should display the credential but clearly indicate that it has been revoked and is no longer valid.	—	Pass
TC-15	Unauthorized Operation	Student or verifier attempts to issue or revoke a credential.	System should deny access and prevent the operation.	—	Pass
TC-16	Boundary Testing	Enter minimum, maximum, below-minimum and above-maximum values for applicable fields such as password/name/year.	Valid boundary values should be accepted and values outside the allowed range should be rejected.	—	Pass
TC-17	Blockchain/Network Failure	Disconnect network or make blockchain unavailable during credential issuance/verification.	System should handle the failure gracefully,	—	Pass

Test Case ID	Test Scenario	Test Steps / Test Data	Expected Result	Actual Result	Pass/Fail
			display an appropriate message, and avoid inconsistent or duplicate records.		
TC-18	Logout and Session Management	Login → logout; separately allow session to expire → access protected page.	User should be logged out and protected resources should require authentication again.	—	Pass
TC-19	Multiple Verification Requests	Send several credential verification requests at the same time.	System should process requests correctly without unacceptable performance degradation.	—	Pass
TC-20	Usability and Compatibility	Use the platform on supported browsers/devices and perform credential verification.	Interface should be readable, understandable, responsive, and functional on supported platforms.	—	Pass

5. Test Design Techniques

The following test design techniques are applied to improve test coverage.

5.1 Equivalence Partitioning

Equivalence Partitioning divides input data into groups that are expected to behave similarly.

Examples:

- **Email:** Valid email / Invalid email

- **Password:** Valid password / Invalid password
- **Credential ID:** Existing ID / Non-existing ID
- **Student ID:** Valid format / Invalid format
- **Graduation Year:** Valid range / Invalid range
- **User Role:** Authorized / Unauthorized
- **QR Code:** Valid QR / Invalid or corrupted QR

For example, instead of testing every possible credential ID, valid and invalid credential IDs can be selected from their respective partitions.

5.2 Boundary Value Analysis

Boundary Value Analysis is applied to fields that have minimum or maximum limits.

Important boundary conditions include:

- Minimum password length
- Maximum password length
- Minimum student name length
- Maximum student name length
- Minimum/maximum permitted graduation year
- Minimum/maximum credential ID length

For each applicable field, the following values should be tested:

Minimum – 1, Minimum, Normal Value, Maximum, Maximum + 1

This helps identify errors occurring at the limits of accepted input.

5.3 Decision Table Testing

Credential verification depends on several conditions:

1. Does the credential exist?
2. Does the stored hash match the credential?
3. Has the credential been revoked?

The resulting decisions are:

- Credential exists + hash matches + not revoked → **Credential Valid**
- Credential exists + hash mismatch → **Credential Tampered/Invalid**
- Credential does not exist → **Credential Not Found**
- Credential exists + revoked → **Credential Revoked**

This ensures that different combinations of verification conditions are tested.

5.4 State Transition Testing

An academic credential can pass through different states during its lifecycle:

Created → Blockchain Stored → Active → Revoked

Important transitions include:

- Credential created → blockchain transaction successful → **Active**
- Active credential → verification request → **Valid**
- Active credential → authorized revocation → **Revoked**
- Revoked credential → verification request → **Revoked**
- Unauthorized user → revocation attempt → **Access Denied**

State Transition Testing ensures that the system responds correctly to actions performed at different stages of the credential lifecycle.

6. Test Coverage

Test Area	Requirements/Tests Covered
Registration & Authentication	FR-01, FR-12; TC-01 to TC-04, TC-18
Credential Issuance	FR-02, FR-03; TC-05, TC-06
Blockchain Storage	FR-04; TC-08, TC-17
Credential ID & QR	FR-05; TC-07
Credential Access	FR-06; TC-09
Credential Verification	FR-07; TC-10, TC-11
Tamper Detection	FR-08; TC-12
Credential Revocation	FR-09, FR-10; TC-13, TC-14
Authorization & Security	FR-11, NFR-01; TC-15
Boundary Testing	TC-16
Exception Handling	NFR-05; TC-17
Performance	NFR-03, NFR-04; TC-19
Usability & Compatibility	NFR-06, NFR-07; TC-20

7. Test Execution Status

The Actual Result and Pass/Fail columns will be filled during actual execution.

- **Pass:** Actual result matches the expected result.
- **Fail:** Actual result differs from the expected result.
- **Blocked:** Test cannot be performed because of an environment or dependency issue.
- **Not Executed:** Test has not yet been performed.

8. Conclusion

The test case design provides comprehensive coverage for the Blockchain-Based Academic Credential Verification Platform. It covers the major functional requirements such as user authentication, credential issuance, blockchain storage, credential verification, QR-based verification, tamper detection, revocation, authorization, and session management. It also considers important non-functional aspects including security, integrity, performance, reliability, usability, and compatibility. The use of Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing ensures that normal, invalid, boundary, and exceptional conditions are adequately tested. Overall, these test cases help ensure that academic credentials are authentic, secure, tamper-resistant, reliable, and easily verifiable.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25